M301 Linear Algebra: Exam Study Guide

Prepared for the October 21 exam

Contents

1 Vectors and linear combinations **1**

2 Matrices and row reduction **2**

3 Linear transformations and determinants **2**

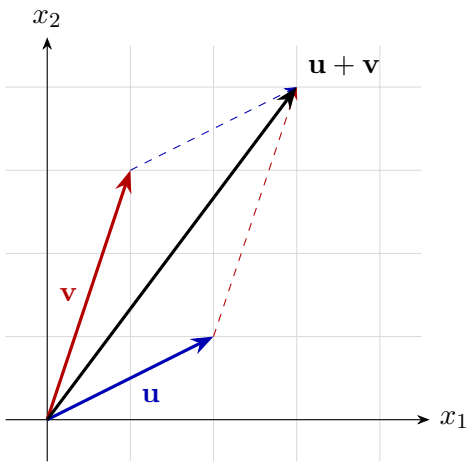
4 Eigenvalues and eigenvectors **2**

5 Practice problems **3**

1 Vectors and linear combinations

A vector in $\mathbb{R}^n$ is an ordered list of n real numbers. Vectors add entry by entry and scale by multiplying every entry:

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \mathbf{u} + \mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad 2\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$



Definition. *The* span *of* $\mathbf{v}_1, \ldots, \mathbf{v}_p$ *is the set of all linear combinations* $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p$. *The vectors are* linearly independent *when* $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$ *only for* $c_1 = \cdots = c_p = 0$.

Check: put the vectors as columns of A and row reduce. They are independent exactly when every column has a pivot.

2 Matrices and row reduction

Solve $A\mathbf{x} = \mathbf{b}$ by row reducing the augmented matrix:

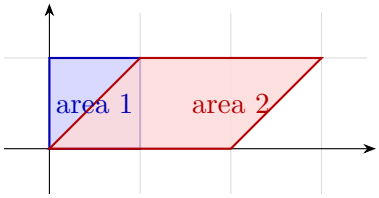
$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 5 & 1 & 8 \\ -1 & -1 & 4 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right], \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}.$$

Row operation	Effect on $\det A$	Example
Swap two rows	changes the sign	$R_1 \leftrightarrow R_2$
Scale a row by k	multiplies by k	$R_2 \to 3R_2$
Add a multiple of a row	no change	$R_3 \to R_3 + 2R_1$

Theorem (Invertible matrix theorem, part)**.** *For an $n \times n$ matrix A, these say the same thing: A is invertible;* $\det A \neq 0$*; $A\mathbf{x} = \mathbf{0}$ has only the trivial solution; the columns of A span $\mathbb{R}^n$; 0 is not an eigenvalue of A.*

3 Linear transformations and determinants

The matrix $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ sends the unit square to a parallelogram whose area is $|\det A| = 2$.

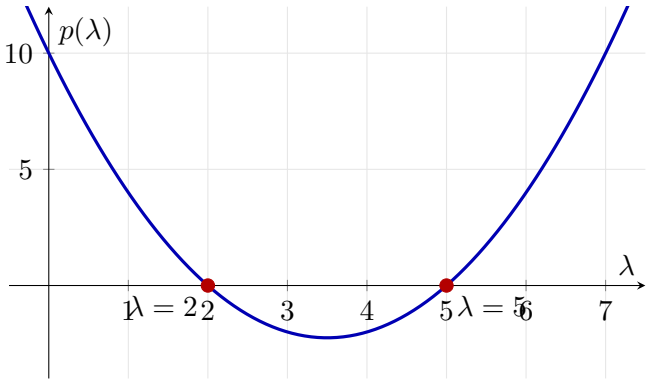


For a 2×2 matrix, $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$. For larger ones, expand along a row or column with many zeros, or row reduce to triangular form and multiply the diagonal.

4 Eigenvalues and eigenvectors

λ is an eigenvalue of A when $A\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \neq \mathbf{0}$; find them as the roots of the characteristic polynomial $\det(A - \lambda I) = 0$. For $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$:

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$



5 Practice problems

1. Are $\begin{bmatrix}1\\2\\3\end{bmatrix}$, $\begin{bmatrix}4\\5\\6\end{bmatrix}$, $\begin{bmatrix}7\\8\\9\end{bmatrix}$ linearly independent? Explain with row reduction.

2. Find $\det \begin{bmatrix}1 & 2 & 0\\3 & 1 & 4\\0 & 2 & 1\end{bmatrix}$.

3. Find the eigenvectors of $A = \begin{bmatrix}4 & 1\\2 & 3\end{bmatrix}$ for $\lambda = 2$ and $\lambda = 5$.

4. Describe what $\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}$ does to the plane, and give its determinant.